



IonQ Quantum Cloud

Platform for accessing trapped-ion quantum computers
Forte & Aria via the cloud.



Trapped-Ion Hardware



Hybrid Workflows



Enterprise Scale

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Why IonQ Quantum Cloud?



The Challenge

Classical computing architectures hit a wall when dealing with complex combinatorial problems. As variables increase, computational difficulty scales exponentially, making certain tasks intractable.

- ✗ Combinatorial optimization limits in logistics
- ✗ Inaccurate molecular simulation for drugs/materials
- ✗ Machine learning bottlenecks at scale



IonQ Purpose

Provide secure, scalable access to high-fidelity trapped-ion quantum computers and simulators. Enable developers to prototype, benchmark, and deploy hybrid quantum-classical workflows today.

Key Applications



Chemistry

Molecular simulation & materials



Optimization

Logistics & portfolio mgmt



ML / AI

Generative models & classification



CAE Design

Structural optimization

Key Features & Capabilities

Enterprise-grade quantum infrastructure powered by trapped ions

IonQ Forte

Latest Generation



Physical Qubits
36

1Q Gate Error
~0.02%

2Q Gate Error
~0.4%

High-performance, software-configurable system featuring acousto-optic deflectors (AODs) for dynamic qubit control.

IonQ Aria

Production Ready



Physical Qubits
25

1Q Gate Error
~0.06%

2Q Gate Error
~0.6%

Reliable workhorse for commercial applications, featuring glass trap technology and high-fidelity operations.

Architecture

- ✓ **All-to-All Connectivity:**
Any qubit can entangle with any other directly.
- ✓ **Long Coherence:**
Natural ion qubits offer seconds of coherence time.

Software Stack

- ✓ **Error Mitigation:**
Built-in debiasing & sharpening to improve results.
- ✓ **Simulators:**
Ideal & noise-model simulators (up to 29 qubits).

Operations

- ✓ **Flexible Access:**
On-demand pay-as-you-go or reserved capacity.
- ✓ **Native Gates:**
Direct access to native gates for optimization.

Sources

ionq.com/quantum-systems/forte • ionq.com/quantum-systems/aria • docs.ionq.com

Developer Experience

Seamless integrations with major quantum SDKs and cloud platforms

Access Channels



IonQ Quantum Cloud
Direct Native Access & Console

Recommended



Amazon Braket
Managed Service



Azure Quantum
Managed Service

SDK Support

● Qiskit

● Cirq

● Q# / QDK

● PennyLane

● TensorFlow Quantum

● TKET

Quickstart

Qiskit Cirq Braket



hello_ionq.py

```
from qiskit_ionq import IonQProvider
from qiskit import QuantumCircuit
```

Sources

docs.ionq.com/sdks • aws.amazon.com/braket • azure.microsoft.com/quantum

Who Benefits & What You Need

Target personas, technical prerequisites, and developer constraints

Target Developer Personas



PRIMARY

Algorithm Developers

Researchers designing novel circuits for chemistry simulations, QAOA, and VQA. Focused on gate fidelity and qubit connectivity.



BUILDER

App Developers

Engineers integrating quantum kernels into classical software stacks via APIs. Focused on stability, SDKs, and hybrid workflows.



STRATEGIC

R&D Teams

Corporate and academic groups validating hardware capabilities against classical baselines. Focused on benchmarking #AQ.



OPS

Enterprise IT

Innovation leads managing cloud quotas, access control, and vendor evaluation. Focused on security and scalability.

Prerequisites



Python Proficiency

Primary language for Qiskit, Cirq, and IonQ SDKs.



Quantum Fundamentals

Understanding of gates, superposition, and measurement.



Cloud Account

Access to IonQ Cloud, AWS Braket, or Azure Quantum.

Dev Profile Constraints

When deploying via Azure Quantum, IonQ targets use the **QIR Base Profile**.

No dynamic control flow (if/else) based on mid-circuit measurement results.

Circuits must be fully defined before submission.

Return type restricted to measurement result histograms.

Sources

docs.ionq.com/getting-started • learn.microsoft.com/azure/quantum

Limitations & Risks

Current hardware constraints and platform considerations



HARDWARE SCALE

Limited Qubit Count

Current systems max out at **36 physical qubits** (Forte) and **25 qubits** (Aria).

- ▶ Insufficient for full fault tolerance or large-scale error correction.
- ▶ Deep circuits accumulate noise; requires error mitigation strategies.



QIR PROFILE

Static Control Flow

Azure Quantum integration uses the **Base Profile**, restricting dynamic capabilities.

- ▶ No mid-circuit measurement-conditioned branching (if/else).
- ▶ Cannot implement iterative phase estimation or adaptive protocols.



OPERATIONS

Throughput & Cost

Shared cloud queues can lead to wait times; pricing is usage-based (QGS).

- ▶ High shot counts for error mitigation increase execution cost.
- ▶ Queue delays can impact iterative variational algorithm loops.



Vendor Specific Ecosystem

While accessible via AWS/Azure, IonQ Cloud is not hardware-agnostic. Code written for IonQ native gates may not be portable to superconducting systems (e.g., Rigetti, IBM) without transpilation.



API Stability (Beta)

API v0.4 is currently in beta. Interfaces, payload structures, and feature sets may change, potentially requiring code updates for production workflows.

Sources

learn.microsoft.com/azure/quantum • docs.ionq.com/api-reference

Roadmap & Future Direction

Aggressive scaling towards fault tolerance and photonic interconnects



Sources

Real-World Results

Industry leaders delivering value with IonQ technology today



AstraZeneca
Life Sciences

20x

Faster Time-to-Solution

Accelerated molecular simulation for drug development using hybrid quantum-classical workflows with AWS & NVIDIA.



Ansys
Engineering

12%

Performance Gain

Outperformed classical solvers in medical device design optimization involving millions of vertices.



Airbus
Aerospace

Largest

Variational Optimization

Utilized IonQ Forte to optimize complex cargo loading configurations, demonstrating scale for logistics.



Hyundai Motor

Automotive AI

Quantum Machine Learning

Applied quantum kernels to image classification and 3D object detection for autonomous driving systems.



GE Research

FinTech

Risk Aggregation

Modeled multi-variable financial risk using copula-based quantum algorithms to predict market behaviors.

Sources

ionq.com/customers-and-partners • ionq.com/case-studies